Day 076 | Unit 06 | Teacher plan

Understand and Use Ratios and Rates

Describe Ratios with Ordered Labels

Standards: 6.RP.A.1

Learning target: I can describe Ratios with Ordered Labels, and explain and check my reasoning.

enVision Topic 5: Understand and Use Ratios and Rates | Spectrum companion Chapter 3

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. A ratio compares two quantities multiplicatively. In a:b, a names the first quantity and b names the second. Divide both terms by a common factor to simplify, while keeping the labels and order. For a:b, the total is a+b equal parts. The first part is a/(a+b) of the total; a/b compares it to the second part. Write labels before choosing the denominator.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 6

Day 076 | Unit 06 | Lesson

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01 / Learn

A ratio compares two quantities multiplicatively. In a:b, a names the first quantity and b names the second. Divide both terms by a common factor to simplify, while keeping the labels and order. For a:b, the total is a+b equal parts. The first part is a/(a+b) of the total; a/b compares it to the second part. Write labels before choosing the denominator.

Worked example 1

A tray has 3 red tiles and 5 blue tiles. Write the red-to-blue ratio in simplest form.

1. A ratio compares two quantities multiplicatively.

2. In a:b, a names the first quantity and b names the second.

3. Keep red first

4. divide 3:5 by 1.

5. State the result in the form requested: 3:5.

Answer: 3:5

Worked example 2

A bag contains 25 red tiles and 30 blue tiles. Select all true statements.
A. The fraction of all tiles that are red is 5/11.
B. The red-to-blue ratio is 5:11.
C. The red-to-blue ratio is 5:6.
D. There are 56 tiles in all.

1. For a:b, the total is a+b equal parts.

2. Divide both color counts by 5.

3. For the fraction of the whole, use total 55 as denominator

4. red-to-blue compares just the two colors.

Answer: A, C - The fraction of all tiles that are red is 5/11.; The red-to-blue ratio is 5:6.

Day 076 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

A tray has 3 red tiles and 2 blue tiles. Write the red-to-blue ratio in simplest form.

02 / Constructed response

A collection has 10 green and 15 yellow counters. Write the fraction that is green.

03 / Constructed response

A tray has 5 red tiles and 4 blue tiles. Write the red-to-blue ratio in simplest form.

04 / Constructed response

A collection has 25 green and 20 yellow counters. Write the fraction that is green.

05 / Constructed response

A tray has 25 red tiles and 25 blue tiles. Write the red-to-blue ratio in simplest form.

06 / Constructed response

A collection has 40 green and 20 yellow counters. Write the fraction that is green.

07 / Constructed response

A tray has 4 red tiles and 12 blue tiles. Write the red-to-blue ratio in simplest form and complete: there are ___ times as many red as blue.

08 / Multiple select

A bag contains 10 red tiles and 15 blue tiles. Select all true statements.
A. The red-to-blue ratio is 2:3.
B. There are 26 tiles in all.
C. The red-to-blue ratio is 2:5.
D. The fraction of all tiles that are red is 2/5.

09 / Multiple select

A bag contains 10 red tiles and 30 blue tiles. Select all true statements.
A. The red-to-blue ratio is 2:8.
B. The red-to-blue ratio is 2:6.
C. The fraction of all tiles that are red is 1/4.
D. There are 41 tiles in all.

10 / Constructed response

A collection has 18 green and 36 yellow counters. Write the fraction that is green and the green-to-yellow ratio in simplest form. Explain the different denominators.

Day 076 | Unit 06 | Answer key

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1. 3:2

Keep red first; divide 3:2 by 1.

6.RP.A.1 | Constructed response

2. 2/5

Total = 25; green fraction = 10/25 = 2/5.

6.RP.A.1 | Constructed response

3. 5:4

Keep red first; divide 5:4 by 1.

6.RP.A.1 | Constructed response

4. 5/9

Total = 45; green fraction = 25/45 = 5/9.

6.RP.A.1 | Constructed response

5. 1:1

Keep red first; divide 25:25 by 25.

6.RP.A.1 | Constructed response

6. 2/3

Total = 60; green fraction = 40/60 = 2/3.

6.RP.A.1 | Constructed response

7. 1:3; 1/3 times as many

Keep red first; divide 4:12 by 4.

6.RP.A.1 | Constructed response

8. A, D - The red-to-blue ratio is 2:3.; The fraction of all tiles that are red is 2/5.

Divide both color counts by 5. For the fraction of the whole, use total 25 as denominator; red-to-blue compares just the two colors.

6.RP.A.1 | Multiple select

9. B, C - The red-to-blue ratio is 2:6.; The fraction of all tiles that are red is 1/4.

Divide both color counts by 5. For the fraction of the whole, use total 40 as denominator; red-to-blue compares just the two colors.

6.RP.A.1 | Multiple select

10. 1/3; 1/2 or 1:2; the whole includes both colors

Total = 54; green fraction = 18/54 = 1/3.

6.RP.A.1 | Constructed response